[INCH-POUND]
A-A-3084
November 7, 1996

COMMERCIAL ITEM DESCRIPTION

PROTRACTORS, MECHANIC'S, BEVEL AND PLAIN, RECTANGULAR HEAD

The General Services Administration has authorized the use of this Commercial Item Description for the procurement of the type III, style B protractor specified in GGG-P-676 for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers protractors designed for precision measuring, checking, laying out of angular work, and scribing layout lines.

2. CLASSIFICATION.

2.1 Classification. Not applicable.

3. SALIENT CHARACTERISTICS.

3.1 Design. The bevel and plain protractor shall have a rectangular head with graduations and a non-adjustable blade. The protractor shall be made of stainless or corrosion resistant steel, heat treated to produce a protractor capable of long life under service conditions. The protractor shall be similar to Figure 1.

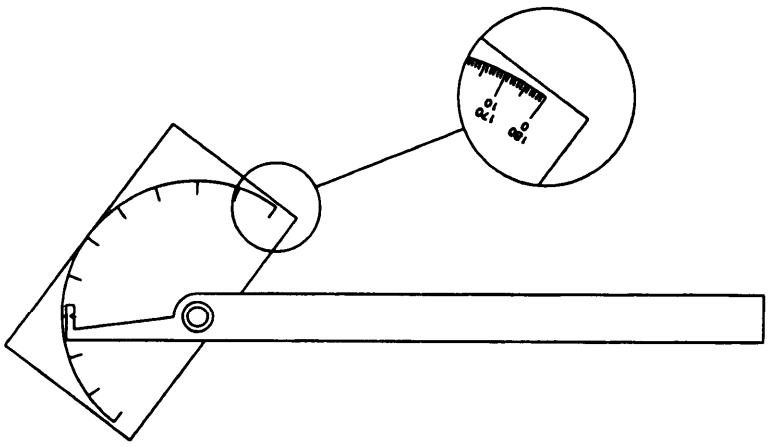


Figure 1. Protractor, plain, rectangular head.

3.1.1 Head. The rectangular head protractor shall be not less than 3-1/4 inches nor more than 3-3/8 inches in length, and not less than 1-7/8 inches nor more than 2 inches in width. The protractor shall have four finished working edges and the surface shall be flat, parallel, and square within 0.005 inch. It shall have two scales

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

Distribution Statement A. Approved for public release; distribution is unlimited.

FSC 5210

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:29Z
Check the source to verify that this is the current version before use.

accurately graduated in degrees, reading both ways to show the complimentary degrees from 0 degree to 180 degrees on each scale, and shall be accurate to within ± 5 minutes between any two graduations. In addition, each graduation marking shall be within 10 minutes of its true reading.

3.1.2 Blade. The indicating arm of the blade shall have a line graduation for accurate setting and reading of the protractor. The length of the blade from the center of the stud hole shall be not less than 6 inches nor more than 6-1/4 inches with a width of 5/8 inch and a thickness of 1/16 inch. The working edges shall be accurately straight within 0.0005 inch. The blade shall be mounted on the head in such a manner as to allow free and accurate positioning.

3.1.3 Thumb nut. A knurled thumb nut or similar locking device shall be furnished to allow for adjusting tension and locking the blade at any position.

3.2 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.3 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:29Z
Check the source to verify that this is the current version before use.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

5.5 Test procedures. Tests for accuracy shall be made with suitable equipment of proven accuracy. All tests shall be performed at 68 degrees F, ± 10 degrees F. All tests shall be performed at a constant temperature.

5.5.1 Flatness. The flatness shall be determined with a test indicator in conjunction with a surface gauge and surface plate.

5.5.2 Squareness. The squareness shall be determined by means of a precision square and gauge blocks.

5.5.3 Angular intervals. Angular intervals shall be determined by means of either a circular measuring machine, a sine bar, or angular blocks to determine the total angular error.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(d) If first article samples are required and the number of samples required.
(e) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 2.5, expressed in terms of defects per hundred units.

(f) What preservation, packing, packaging, and marking of packages are required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:29Z
Check the source to verify that this is the current version before use.

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the protractors specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN

5210-00-390-5610

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:29Z
Check the source to verify that this is the current version before use.